DB-Fusion: Dual-Balanced Sensor Fusion for MultiModal 3D Object Detection

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Abstract. Fusing LiDAR and camera features in Bird's Eye View (BEV) space has become the dominant paradigm for multimodal 3D object detection. However, BEV fusion inherently introduces two fundamental yet overlooked structural imbalances that degrade detection quality. **Modality-Occupancy Imbalance (MOI)** allows dense image features to suppress sparse but geometrically precise LiDAR cues, while **Semantic-Occupancy Imbalance (SOI)** causes dominant background regions to dilute class-specific foreground signals. We propose **D**ual-**B**alanced Fusion (**DB-Fusion**), a unified framework that jointly mitigates both. For MOI, Modality-Balanced Feature Sampling closes the cross-modal density gap with depth-guided pseudo-points, and Geometric-based Feature Fusion adaptively weights each modality according to distance- and angle-dependent reliability. For SOI, Semantic-Balanced Feature Enhancing partitions the BEV map into semantic classes and performs class-consistent context aggregation through instance-scene interaction. Extensive experiments on nuScenes confirm that both imbalances constitute practical bottlenecks: addressing them jointly improves mAP to 73.2 on the validation set and 73.6 on the test set, with competitive NDS, with the largest gains emerging from their combination.

Keywords: LiDAR-Camera Fusion · 3D Detection · Bird's-Eye-View

1 Introduction

Fusing LiDAR and camera features in Bird's Eye View (BEV) space has become the standard paradigm for multimodal 3D object detection [17, 20, 23, 30, 35, 37]. By projecting both modalities onto a unified top-down space, BEV enables natural cross-modal alignment of LiDAR's geometric precision and camera's semantic richness. While this paradigm has driven substantial progress, we observe that the fusion process itself introduces two fundamental yet overlooked structural imbalances that consistently degrade detection quality. These imbalances—one between modalities, the other between foreground and background—co-occur in every BEV-based fusion pipeline.

As illustrated in Fig. 1, these imbalances take two forms. *(i)* **Modality-Occupancy Imbalance (MOI)** arises because LSS-style lifting [24] produces dense image features across every BEV cell, while LiDAR features remain inherently sparse, particularly at long range. This density gap allows the sheer

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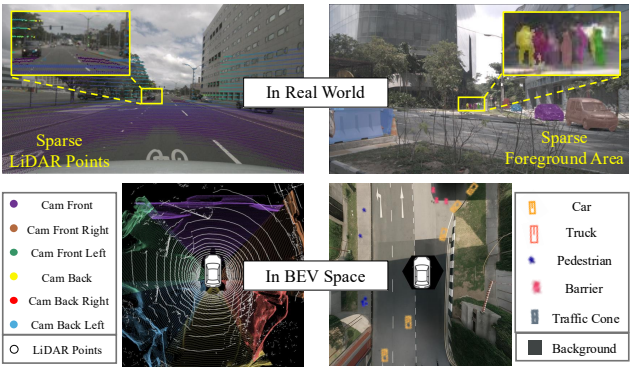


(a) Modality Occupancy Imbalance (b) Semantic Occupancy Imbalance

Fig. 1: Illustration of two major imbalance types in LiDAR–Camera fusion. (a) Modality–Occupancy Imbalance (MOI): *Top*: LiDAR points showing severe long-range sparsity. *Bottom*: BEV-projected camera features revealing the density gap with LiDAR. **(b) Semantic–Occupancy Imbalance (SOI):** *Top*: foreground semantics (e.g., pedestrian, barrier) occupy only a small area. *Bottom*: BEV view showing background dominance and weakened foreground cues.

volume of camera signals to suppress geometrically precise LiDAR cues, biasing the model toward the dense image branch. *(ii)* **Semantic-Occupancy Imbalance (SOI)** stems from the severe spatial disparity between foreground and background in BEV: objects of interest occupy only a small fraction of grid cells, while the surrounding background dominates. Recovering such sparse objects requires gathering class-specific semantic cues, yet existing methods [1,30,37] often aggregate context indiscriminately across the entire scene, letting cross-class interference dilute the class-specific cues—road structure for vehicles, sidewalks for pedestrians. Critically, the two imbalances are not independent but tightly coupled. In LiDAR-sparse regions, dense image cues can provide useful support, but without proper modality balancing, they may entangle weak object cues with dominant background semantics. Indiscriminate scene-level context aggregation then compounds this problem by incorporating irrelevant semantic context rather than selectively reinforcing foreground signals, making class-specific information harder to preserve. Addressing each imbalance separately has inherent limitations; the full benefit is achieved only when both are handled jointly.

Unlike LSS-style lifting [24], projection-based methods [4,26] avoid modality density mismatch by sampling image features only at LiDAR-occupied locations, but discard valuable camera information in LiDAR-sparse regions (e.g., long range). IS-Fusion [37] partially alleviates foreground-background disparity through instance-scene interaction, yet its class-agnostic aggregation still introduces cross-class interference. Query-based methods [1, 12] sidestep dense BEV fusion by decoding objects directly from sparse queries, but rely primarily on local context and thus underutilize broader scene information. Overall, prior work operates along a single axis—either reducing modality density mismatch or en-

hancing foreground emphasis—while the two problems fundamentally co-occur and compound each other.

To address these limitations, we propose **D**ual-**B**alanced Fusion (**DB-Fusion**), a framework that jointly mitigates MOI and SOI within a single pipeline. The key insight is that the two imbalances are structurally coupled—addressing only one leaves the other unresolved—and thus demand a unified solution. To mitigate MOI, we introduce **Modality-Balanced Feature Sampling (Mo-Bal)**, which supplements sparse LiDAR-guided image sampling with depth-predicted pseudo-points to close the density gap. **Geometric-based Feature Fusion (GeoFuse)** then models distance- and angle-dependent sensor reliability to adaptively weight each modality per grid cell. To resolve SOI, we propose **Semantic-Balanced Feature Enhancing (Se-Bal)**, which partitions the BEV map into semantic classes and performs class-consistent context aggregation with instance-scene interaction, ensuring that each object draws contextual support from its semantically relevant surroundings rather than the entire scene.

Extensive experiments on the nuScenes benchmark [2] confirm that both imbalances have a measurable impact on detection performance: starting from a 70.7 mAP baseline, mitigating MOI alone improves mAP by +1.6, mitigating SOI alone by +2.1, and addressing both jointly yields 73.2 mAP with competitive NDS—with the largest gains emerging from their combination. DB-Fusion further demonstrates strong generalization on the nuScenes test set, reaching 73.6 mAP. Detailed ablations, attention visualizations, and failure-case comparisons further demonstrate that each module targets its intended imbalance effectively. Beyond the resulting accuracy, the central contribution of this work is a principled framework that identifies MOI and SOI as the two structural bottlenecks of BEV-based fusion and resolves both within a single pipeline.

2 Related Works

2.1 BEV Space Generation

Generating Bird's-Eye-View (BEV) representations has become fundamental for unified multimodal perception. LSS [24] pioneered explicit depth prediction to lift image features into 3D space, with subsequent works [11, 18] refining depth supervision strategies. However, these lifting-based methods produce uniform feature density across the entire BEV space, where every grid cell receives camera features regardless of actual object presence or LiDAR coverage. BEVFusion [23] observed that this leads to imbalanced fusion, as the sheer volume of image features can suppress sparser but geometrically accurate LiDAR signals. Despite this observation, BEVFusion still lifts entire images uniformly without adapting to local LiDAR distribution. An alternative approach directly projects LiDAR points onto images and samples features only at these locations [4,26,37]. While this avoids density imbalance, it creates the opposite issue by only utilizing camera information where LiDAR points exist, thereby ignoring potentially valuable image features in sparse or occluded regions. Existing approaches thus operate at two extremes: they are either uniformly dense or strictly sparse. More

critically, they lack explicit mechanisms to adaptively balance feature density according to local LiDAR coverage or to assess complementary modality reliability across different spatial contexts.

2.2 LiDAR-Camera Fusion Models

Multimodal fusion leverages complementary strengths of LiDAR geometry and camera semantics. We analyze existing approaches through two key challenges: how to balance modality-specific feature density during fusion, and how to prevent background features from overwhelming foreground signals in BEV space.
BEV-Level Fusion: Recent methods perform fusion in unified BEV space to align heterogeneous sensor data. TransFusion [1] initializes object queries from both modalities, while BEVFusion [20, 23] constructs separate BEV representations and fuses them via concatenation or convolution. However, these approaches apply uniform fusion operations across all BEV locations without considering varying modality reliability. Furthermore, scene-level fusion [25] treats all BEV regions equally, allowing background areas that occupy majority space to dilute spatially sparse foreground signals.
Advanced Fusion Strategies: More recent works have explored refined mechanisms. SparseFusion [31] and UniTR [29] adopt sparse query architectures for efficiency, but do not adjust their fusion or sampling strategies based on whether LiDAR points are abundant or sparse in each spatial region. IS-Fusion [37] introduces instance-level aggregation to enhance local context around objects, representing progress toward addressing background dominance. However, its global attention allows cross-class interference, and point-centric sampling leaves valuable image information unexploited in LiDAR-sparse regions. SparseLIF [40] performs instance-level fusion but lacks geometric constraint modeling for context-dependent reliability. MambaFusion [30] enables efficient cross-modal exchange yet fuses at scene level without semantic-aware routing.

Existing methods reveal consistent limitations. BEV-level methods apply uniform fusion without adapting to geometric context or semantic distinctions. Advanced methods focus on specific aspects such as computational efficiency and foreground emphasis. However, none jointly address both modality density imbalance and semantic feature dilution within a unified framework.

3 Methodology

3.1 Overview

As illustrated in Fig. 2, our DB-Fusion framework processes LiDAR point clouds $\mathcal{P}$ and multi-view images $\{I^v\}_{v=1}^{N_{\text{cam}}}$ through modality-specific encoders [22, 41] to obtain LiDAR features F_{L} and image features F_{I}. The LiDAR BEV feature $B_{\text{L}} \in \mathbb{R}^{W \times H \times C}$ is obtained by compressing the height dimension of F_{L}, where W and H denote the spatial dimensions, and C denotes the channel dimension of the BEV grid. For the image modality, we first apply LiDAR-to-Camera

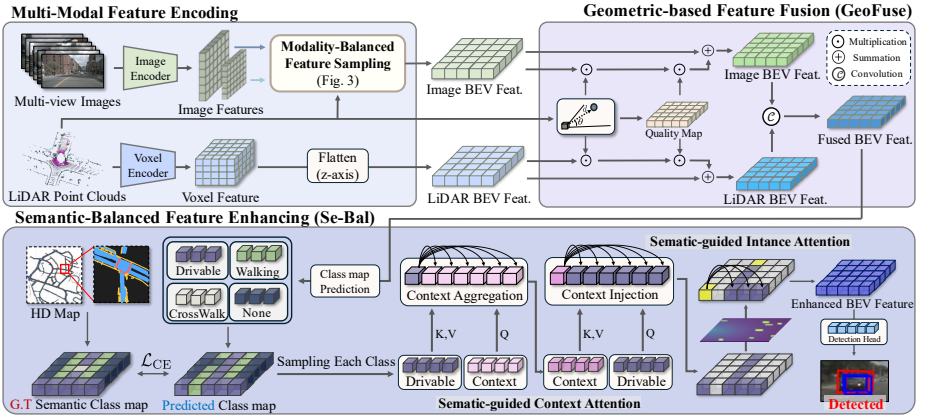


Fig. 2: Overview of DB-Fusion. Multi-view images and LiDAR point clouds are encoded into BEV features. Mo-Bal densifies image BEV features with depth-guided pseudo points, GeoFuse adaptively fuses LiDAR and image BEV features using geometry-derived reliability, and Se-Bal enhances the fused BEV features through semantic-guided contextual aggregation and instance-scene interaction.

projection [37] to obtain anchor image BEV features B_{anchor} at LiDAR point locations. To supplement sparse LiDAR coverage, **Modality-Balanced Feature Sampling** (**Mo-Bal**, Sec. 3.2) estimates image-based depth, generates pseudo points at plausible locations, and samples additional camera features to obtain augmented image BEV features B_{aug}. The final image BEV feature B_{I} is obtained by fusing B_{anchor} and B_{aug} through a convolutional layer.

Next, **Geometric-based Feature Fusion** (**GeoFuse**, Sec. 3.3) models modality reliability based on geometric context (range, orientation) and learns grid-wise fusion weights to adaptively enhance each modality's features, producing refined features $\hat{B}_{\text{L}}$ and $\hat{B}_{\text{I}}$. These refined features are then concatenated and fused through a convolutional layer to obtain the fused BEV representation B_{F}.

Subsequently, **Semantic-Balanced Feature Enhancing** (**Se-Bal**, Sec. 3.4) partitions the BEV grid B_{F} into semantic classes using an auxiliary segmentation head, aggregates contextual information within each class, and performs class-wise instance-scene interaction, producing the final enhanced BEV representation $\hat{B}_{\text{F}} \in \mathbb{R}^{W \times H \times C}$.

Decoder and Training. The enhanced BEV features $\hat{B}_{\text{F}}$ are fed into a transformer-based decoder [36] that predicts final 3D bounding boxes. Following standard practices, we apply the Hungarian algorithm [13] for matching predictions to ground truth during training. The framework is trained end-to-end with Focal loss [21] for classification, L1 loss for bounding box regression, and auxiliary losses including Smooth L1 loss [8] for depth prediction in Sec. 3.2 and Cross-Entropy loss for BEV segmentation in Sec. 3.4.

3.2 Modality-Balanced feature sampling (Mo-Bal)

The core idea of Mo-Bal is to enrich sparse LiDAR-guided sampling with informative image features while preventing dense image features from overwhelming sparse LiDAR signals. As illustrated in Fig. 3, given LiDAR point cloud $\mathcal{P}$ and image features F_{I}, Mo-Bal constructs an image BEV feature B_{I} by balancing reliable geometric information with dense semantic cues:

$$B_{\mathrm{I}} = f_{\mathrm{fuse}}(B_{\mathrm{anchor}}(\mathcal{P}), B_{\mathrm{aug}}(\mathcal{P}_{\mathrm{aug}})) \tag{1}$$

where B_{anchor} denotes image BEV features sampled at original LiDAR locations $\mathcal{P}$, B_{aug} represents augmented BEV features sampled at pseudo point locations $\mathcal{P}_{\mathrm{aug}}$, and $f_{\mathrm{fuse}}(\cdot)$ is the depth-guided gating fusion function. The key challenge is to generate $\mathcal{P}_{\mathrm{aug}}$ that maintains cross-modal consistency while supplementing sparse LiDAR coverage.

Depth prediction with sparse supervision. To generate $\mathcal{P}_{\mathrm{aug}}$, Mo-Bal employs depth estimation to identify geometrically plausible 3D locations. A lightweight depth prediction network $f_{\mathrm{depth}}(\cdot)$ takes image features F_{I}^{v} as input and outputs per-pixel depth for each camera view v:

$$\hat{D}^{v} = f_{\mathrm{depth}}(F_{\mathrm{I}}^{v}) \in \mathbb{R}^{H_{\mathrm{img}} \times W_{\mathrm{img}}} \tag{2}$$

The network is trained with sparse LiDAR depth supervision using Smooth L1 loss, producing depth predictions $\{\hat{D}^{v}\}_{v=1}^{N_{\mathrm{cam}}}$ across all camera views.

Occlusion-Aware Gating. Using the predicted depth maps $\{\hat{D}^{v}\}_{v=1}^{N_{\mathrm{cam}}}$, Mo-Bal determines which empty BEV cells are suitable for pseudo point generation. For each empty BEV cell at grid location (x, y), we generate a candidate point $p_{\mathrm{cand}} = (x, y, z_0)$ at ground height z_0 and project it to all camera views. The projection to view v yields a pixel location u^{v} and the ground-truth depth d_{cam}^{v} (the depth of p_{cand} in camera v's coordinate system). We then compare d_{cam}^{v} with the network-predicted depth $\hat{D}^{v}(u^{v})$ at pixel u^{v}. A cell is marked as visible if the absolute difference is within an adaptive threshold for at least one view:

$$\mathcal{M}_{\mathrm{vis}}(x, y) = \begin{cases} 1, & \text{if } V(x, y) \neq \emptyset \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $V(x, y) = \{v : |\hat{D}^{v}(u^{v}) - d_{\mathrm{cam}}^{v}| \leq \tau_{\mathrm{abs}} + \tau_{\alpha} \cdot d_{\mathrm{cam}}^{v}\}$ is the set of views from which the cell is visible. The distance-adaptive threshold $\tau_{\mathrm{abs}} + \tau_{\alpha} \cdot d_{\mathrm{cam}}^{v}$ accounts for increasing depth uncertainty at longer ranges.

Based on $\mathcal{M}_{\mathrm{vis}}$, Mo-Bal generates pseudo points only at visible locations. Each pseudo point is sampled around the cell center using truncated normal distributions in XY (clipped within cell bounds) and a normal distribution centered at z_0 in Z, producing the augmented point set $\mathcal{P}_{\mathrm{aug}}$.

Dual-Stream Gating Fusion. Following [37], we obtain $B_{\mathrm{anchor}} \in \mathbb{R}^{W \times H \times C}$ and $B_{\mathrm{aug}} \in \mathbb{R}^{W \times H \times C}$ by sampling image features at $\mathcal{P}$ and $\mathcal{P}_{\mathrm{aug}}$, respectively. The fusion function $f_{\mathrm{fuse}}(\cdot)$ in Eq. (1) is realized as $f_{\mathrm{fuse}} = f_{\mathrm{refine}} \circ f_{\mathrm{gate}}$, where f_{gate} computes adaptive weights and f_{refine} produces the final fused features.

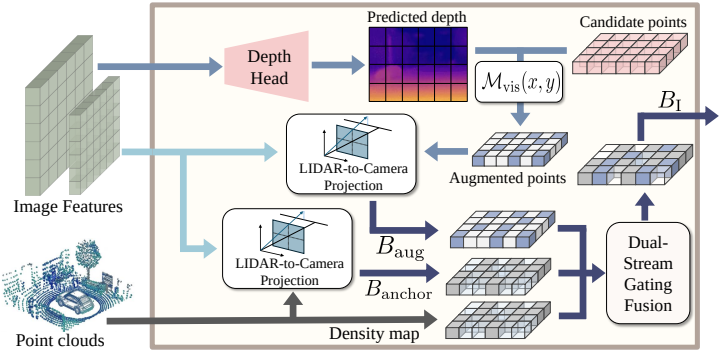


Fig. 3: Illustration of Mo-Bal. Depth prediction identifies visible regions, enabling pseudo point generation at unoccluded areas. Dual-stream gating fusion adaptively combines anchor and augmented BEV features based on LiDAR density.

Specifically, we first construct a LiDAR density map $D_{\text{L}} \in [0,1]^{W \times H}$ from the original point cloud $\mathcal{P}$ by normalizing point counts per BEV cell. The fusion is then performed by:

$$G = \sigma(f_{\text{gate}}([B_{\text{anchor}}, B_{\text{aug}}, D_{\text{L}}])), \tag{4}$$

$$B_{\text{I}} = f_{\text{refine}}(G \odot B_{\text{aug}} + (1 - G) \odot B_{\text{anchor}}) \tag{5}$$

where $[\cdot, \cdot, \cdot]$ denotes the concatenation, $f_{\text{gate}}(\cdot)$ is a convolutional layer producing the gating map $G \in [0,1]^{W \times H \times 1}$, $f_{\text{refine}}(\cdot)$ consists of convolutional layers with batch normalization and ReLU activations, and $\odot$ denotes element-wise multiplication.

3.3 Geometric-based Feature Fusion (GeoFuse)

Given B_{L} and B_{I} from the previous stage, a straightforward approach is to fuse them directly through convolutional networks to obtain the fused BEV feature B_{F}. However, this ignores the distinct geometric characteristics of each sensor: LiDAR exhibits high accuracy at close range but degrades at longer distances [14, 33] and varying elevation angles [6, 10, 34] due to sparsity and measurement uncertainty. In contrast, cameras provide consistent information regardless of distance or viewing angle. By explicitly accounting for these complementary characteristics, we can obtain a more enhanced fused feature B_{F}. Specifically, we compute distance-dependent sensor reliability and angle-based observation quality from geometric cues, generating confidence maps that adaptively enhance B_{L} and B_{I} before fusion.

Distance-based reliability estimation. For each LiDAR point $p_i = (x_i, y_i, z_i)$, we compute the normalized distance $\hat{d}_i = d_i / \max(d_i)$, where $d_i = \sqrt{x_i^2 + y_i^2 + z_i^2}$. Sensor-specific reliability weights are calculated through:

$$w_{\text{L}}(i) = \sigma(-\alpha \cdot (\hat{d}_i - \beta)), \quad w_{\text{I}}(i) = 1 - w_{\text{L}}(i) \tag{6}$$

where $\sigma(\cdot)$ is the sigmoid function, and α, β are learnable parameters controlling the transition slope and center. This ensures complementary weighting with $w_{\mathrm{L}} + w_{\mathrm{I}} = 1$, reflecting that LiDAR reliability decreases while camera reliability increases with distance.

Angle-based quality modeling. LiDAR reflection quality degrades at steep elevation angles. For each point, we compute the elevation angle $\theta_i = \arctan\big(|z_i|/\sqrt{x_i^2 + y_i^2}\big)$ and evaluate observation quality as:

$$q(i) = \exp\left(-\frac{\theta_i}{\gamma \cdot (1 + 0.5 \cdot \hat{d}_i)}\right) \tag{7}$$

where γ is a learnable parameter that controls the angular decay rate. Points near the horizontal plane ($\theta_i \approx 0$) receive higher quality scores, while the distance-adaptive term $(1 + 0.5 \cdot \hat{d}_i)$ relaxes the constraint for nearby objects where steep angles are more common.

BEV Mapping and Feature refinement. The point-wise reliability weights $w_{\mathrm{L}}(i), w_{\mathrm{I}}(i)$ and angle quality scores $q(i)$ are mapped to BEV grids by converting points to grid coordinates and averaging values from multiple points per cell, producing BEV confidence maps $M_{\mathrm{L}}, M_{\mathrm{I}}, M_{\mathrm{quality}} \in \mathbb{R}^{W \times H}$. Using these maps, we enhance each modality's features:

$$\hat{B}_m = B_m \odot (1 + M_{\mathrm{quality}} \odot M_m), \quad m \in \{\mathrm{L}, \mathrm{I}\} \tag{8}$$

The enhanced features $\hat{B}_{\mathrm{L}}$ and $\hat{B}_{\mathrm{I}}$ are then fused via a 3×3 convolution to obtain B_{F}, adaptively leveraging each sensor's strengths according to distance and observation geometry.

3.4 Semantic-Balanced Feature Enhancing (Se-Bal)

With the fused BEV feature B_{F}, global attention is commonly used for context aggregation. However, foreground objects have distinct class identities (e.g., vehicles, pedestrians) and benefit from interacting with semantically relevant context rather than the entire scene indiscriminately. For instance, vehicle detection benefits from road structure context, while pedestrian detection relies on sidewalk and crosswalk cues. To capture such semantic structure, we construct a semantic BEV map by rasterizing the nuScenes High Definition (HD) map [2] around the ego pose into a BEV grid aligned with B_{F}, where map elements are grouped into semantic classes. Se-Bal leverages this semantic partitioning by aggregating context within each class independently, enabling foreground objects to interact only with their semantically relevant regions.

Semantic routing via BEV segmentation. To partition the BEV grid into semantic classes, we employ an auxiliary segmentation head $f_{\mathrm{seg}}(\cdot)$ that predicts class probabilities over $\mathcal{S} = \{\textit{drivable}, \textit{walkway}, \textit{crosswalk}, \textit{none}\}$:

$$P = f_{\mathrm{seg}}(B_{\mathrm{F}}) \in [0, 1]^{|\mathcal{S}| \times W \times H} \tag{9}$$

Each BEV cell is hard-assigned to the class with maximum probability in P, producing binary masks $\mathcal{M}_s \in \{0,1\}^{W\times H}$ for each class $s \in \mathcal{S}$. These masks partition the BEV feature into class-specific features:

$$B_s = B_{\mathrm{F}} \odot \mathcal{M}_s \in \mathbb{R}^{C\times W\times H}, \quad s \in \mathcal{S} \tag{10}$$

This four-class discretization explicitly models road-related regions where objects predominantly appear, with road markings (e.g., lane dividers, stop lines) grouped into the drivable class.

Semantic-guided context attention. To aggregate context within each semantic class while avoiding cross-class interference, we compress class-specific features into compact context vectors to extract shared information and reduce noise, injecting them back only to corresponding regions.

Aggregation. For each class $s \in \mathcal{S}$, we flatten the class-specific BEV feature B_s into tokens $X_s = \mathrm{Flatten}(B_s) \in \mathbb{R}^{(WH)\times C}$ and introduce a query bank $Q_s \in \mathbb{R}^{N\times C}$ (orthogonally initialized). Class-wise context is aggregated via multi-head cross-attention $f_{\mathrm{MCA}}(\cdot)$:

$$\tilde{Q}_s = f_{\mathrm{MCA}}(Q_s, X_s, X_s) \in \mathbb{R}^{N\times C} \tag{11}$$

where $f_{\mathrm{MCA}}(Q, K, V)$ denotes multi-head cross-attention with query Q, key K, and value V. For the *crosswalk* class, which lies at the boundary between roads and sidewalks, we aggregate context from both *drivable* and *walkway* regions and blend them with a learnable weight $\alpha \in [0,1]$ to capture mixed semantic cues.

Injection. The aggregated context $\tilde{Q}_s$ is injected back only to the corresponding class region:

$$\tilde{X}_s = f_{\mathrm{MCA}}(X_s, \tilde{Q}_s, \tilde{Q}_s) \in \mathbb{R}^{(WH)\times C} \tag{12}$$

After reshaping $\tilde{X}_s$ back to spatial dimensions, we update the class feature with residual gating. This process is applied to all classes except *none*. Finally, the enhanced class-specific features are merged:

$$\tilde{B}_{\mathrm{F}} = \sum_{s\in\mathcal{S}} B_s \in \mathbb{R}^{C\times W\times H} \tag{13}$$

Semantic-guided instance attention. To further enhance the BEV feature with instance-level information, we first extract instance candidates from $\tilde{B}_{\mathrm{F}}$. A heatmap prediction head generates a spatial confidence map, from which we select the top-K peak coordinates and gather their features to form an instance token set $I \in \mathbb{R}^{K\times C}$. Each instance token is assigned to a semantic class based on the maximum probability at its location. Tokens within the same class are grouped, and multi-head self-attention $f_{\mathrm{MSA}}(\cdot)$ is applied within each group:

$$\tilde{I}_s = f_{\mathrm{MSA}}(I_s), \quad s \in \{\textit{drivable}, \textit{walkway}, \textit{crosswalk}\} \tag{14}$$

This allows same-class instances to cross-reference one another, producing refined instance representations. These refined tokens are then injected back

to their corresponding class regions via cross-attention, where the class-specific BEV feature B_s serves as queries:

$$B_s \leftarrow B_s + \eta_s \cdot (f_{\mathrm{MCA}}(B_s, \tilde{I}_s, \tilde{I}_s) \odot \mathcal{M}_s) \tag{15}$$

where $\eta_s \in [0, 1]$ is a learnable gating parameter, and $\mathcal{M}_s$ spatially restricts the update to the corresponding class region. After updating all class-specific features, we reconstruct the final enhanced BEV feature:

$$\hat{B}_{\mathrm{F}} = \sum_{s \in \mathcal{S}} B_s \in \mathbb{R}^{C \times W \times H} \tag{16}$$

This final BEV feature $\hat{B}_{\mathrm{F}}$ is then fed into the detection decoder to produce 3D bounding box predictions, benefiting from both class-consistent context aggregation and instance-scene interaction that sharpen object evidence while suppressing irrelevant background interference.

4 Experiments

4.1 Experiment setup

Dataset and Evaluation. We evaluate our proposed framework **DB-Fusion** on the nuScenes [2] dataset, a widely used large-scale benchmark for multimodal 3D object detection in autonomous driving. The dataset contains 1,000 driving scenes collected in Boston and Singapore. Each scene provides a full 360° field of viewusing six surround cameras and one 32-beam LiDAR sensor. For robustness analysis in Sec. 4.3, we also evaluate on nuScenes-C [7], which applies camera-only corruptions at five severity levels while keeping LiDAR inputs unchanged. For evaluation, we report results using the mean Average Precision (mAP) and the nuScenes Detection Score (NDS) which evaluates object translation, scale, orientation, velocity and attribute. All experiments are conducted on the nuScenes validation set unless otherwise specified.

Training Details. We use a Swin Transformer pretrained on nuImages as the image encoder and VoxelNet [41] as the LiDAR voxel encoder. For detection, we adopt the TransFusion-L head [1], which also serves as our LiDAR-only baseline. To establish a LiDAR–Camera fusion baseline, we incorporate the Point-to-Grid (P2G) module from IS-Fusion [37] into TransFusion-L and denote the resulting model as Baseline-LC. Following the standard nuScenes setting, we use a voxel size of $[0.075, 0.075, 0.2]$ and a point cloud range spanning a 108 m $\times$ 108 m BEV area, yielding a 180 $\times$ 180 BEV feature map. Input images are resized to $(384, 1056)$, and the Swin encoder remains frozen throughout training. In Se-Bal, the segmentation head f_{seg} is supervised with a cross-entropy loss. For class-consistent instance–scene interaction, we retain the top-200 proposals for each frame. All models are trained on 8 RTX 3090 GPUs with a batch size of 2 per GPU.

Table 1: Comparison with recent LiDAR–Camera fusion methods on the nuScenes benchmark. All models utilize both LiDAR and Camera modalities, and all compared fusion frameworks adopt **VoxelNet** [41] as the LiDAR backbone.

Methods	Venue	Mod.	Camera Backbone	*validation set*		*test set*		FPS
				mAP	NDS	mAP	NDS	
PointPainting [26]	CVPR'20	L+C	–	65.8	69.6	–	–	–
PointAugmenting [28]	CVPR'21	L+C	ResNet-101 [9]	–	–	66.8	71.0	–
MVP [38]	NeurIPS'21	L+C	DLA-34 [39]	66.1	70.0	66.4	70.5	–
TransFusion [1]	CVPR'22	L+C	ResNet-50 [9]	67.5	71.3	68.9	71.6	–
AutoAlignV2 [5]	ECCV'22	L+C	CSPNet [27]	67.1	71.2	68.4	72.4	–
UVTR [16]	NeurIPS'22	L+C	ResNet-101 [9]	65.4	70.2	67.1	71.1	–
BEVFusion (PKU) [20]	NeurIPS'22	L+C	Dual-Swin-T [19]	67.9	71.0	69.2	71.8	0.8
DeepInteraction [35]	NeurIPS'22	L+C	ResNet-50 [9]	69.9	72.6	70.8	73.4	2.6
BEVFusion (MIT) [23]	ICRA'23	L+C	Swin-T [22]	68.5	71.4	70.2	72.9	4.2
CMT [32]	ICCV'23	L+C	ResNet-50 [9]	70.3	72.9	72.0	74.1	2.8
SparseFusion [31]	ICCV'23	L+C	ResNet-50 [9]	70.4	72.8	72.0	73.8	5.3
ObjectFusion [3]	ICCV'23	L+C	Swin-T [22]	69.8	72.3	71.0	73.3	–
UniTR [29]	ICCV'23	L+C	Swin-T [22]	70.5	73.3	70.9	74.5	4.9
DAL [12]	ECCV'24	L+C	ResNet-50 [9]	71.5	74.0	72.0	74.8	4.3
SparseLIF [40]	ECCV'24	L+C	V2-99 [15]	71.2	74.6	–	–	2.9
IS-Fusion [37]	CVPR'24	L+C	Swin-T [22]	72.8	74.0	73.0	75.2	3.2
MambaFusion [30]	ICCV'25	L+C	Swin-T [22]	72.7	**75.0**	73.2	**75.9**	4.7
CoreNet [17]	INFFUS'25	L+C	Swin-T [22]	72.5	74.6	73.3	75.6	4.5
DB-Fusion (Ours)	-	L+C	Swin-T [22]	**73.2**	74.5	**73.6**	75.0	3.5

4.2 Benchmark Results

As shown in Tab. 1, DB-Fusion achieves 73.2 mAP on the nuScenes validation set and 73.6 mAP on the test set, placing it among the top-performing LiDAR–camera fusion methods while maintaining an inference speed of 3.5 FPS comparable to other Swin-T-based approaches. Notably, the gain of DB-Fusion is concentrated in mAP rather than NDS. This is because our framework primarily recovers objects that previous methods miss entirely, rather than refining the localization of already-detected objects. Newly recovered detections—often at long range or from rare classes—tend to have less precise orientation and velocity estimates, which limits improvement on the TP-quality terms that dominate NDS. This pattern is consistent with our design goal of reducing the false negatives caused by MOI and SOI, and we provide supporting evidence in Sec. 4.3 and Sec. 4.4.

4.3 Analysis of Imbalance Bottlenecks

Analysis of MOI. We analyze whether MOI is a practical bottleneck by examining how camera bias affects fusion models. On nuScenes-C [7] (Tab. 2), where three corruption types are applied only to camera inputs (averaged over five severity levels) while LiDAR remains unchanged, DB-Fusion consistently achieves the highest mAP and NDS across all corruption types, outperforming IS-Fusion by up to +2.6 mAP and MambaFusion by +1.9 mAP under Gaussian noise. Notably, IS-Fusion samples image features only at LiDAR locations, limiting camera utilization, while MambaFusion lifts full image features into BEV, maximizing it. That both extremes degrade more than DB-Fusion under camera

corruption confirms that neither discarding nor fully embracing image density is sufficient—balancing the two modalities so that neither dominates is key to robust fusion.

Table 2: MOI analysis under camera-only corruptions.

Method	Gaussian		Uniform		Impulse	
	mAP	NDS	mAP	NDS	mAP	NDS
IS-Fusion [37]	65.68	69.99	68.29	71.39	65.67	69.95
MambaFusion [30]	66.37	71.44	68.18	72.44	66.05	71.24
DB-Fusion (Ours)	**68.24**	**71.89**	**70.03**	**72.84**	**68.25**	**71.89**

Analysis of SOI. We examine whether SOI leads to missed detections by analyzing false-negative (FN) counts. Tab. 3(a) reports FN under varying center-distance matching thresholds τ. DB-Fusion consistently achieves the lowest FN count at every threshold, with the gap over MambaFusion widening from +242 at τ=0.5m to +572 at τ=4.0m. This widening gap indicates that DB-Fusion's advantage extends beyond tightly localized objects to loosely matched ones, where background interference is the primary cause of missed detections. Tab. 3(b) further breaks down FN by range. The reduction over IS-Fusion grows from +50 at close range (∼10m) to +172 at long range (∼50m), confirming that class-specific context becomes increasingly critical as raw physical signals weaken with distance. These results validate that SOI is a practical source of missed detections and that DB-Fusion effectively mitigates its impact.

Table 3: False-negative analysis showing the effect of **SOI**.

Method	$\tau = 0.5$m↓	$\tau = 1.0$m↓	$\tau = 2.0$m↓	$\tau = 4.0$m↓
DB-Fusion	**15,818**	**8,305**	**5,056**	**3,722**
IS-Fusion	16,235 (+417)	8,496 (+191)	5,063 (+7)	3,727 (+5)
MambaFusion	16,060 (+242)	8,664 (+359)	5,566 (+510)	4,294 (+572)

(a) FN under different matching thresholds (τ).

Range	DB-Fusion	IS-Fusion
∼10m	**572**	622 (+50)
∼20m	**2,597**	2,634 (+37)
∼30m	**4,226**	4,318 (+92)
∼40m	**4,247**	4,313 (+66)
∼50m	**4,176**	4,348 (+172)

(b) FN by range at $\tau = 0.5$m.

Table 4: Ablation study across Dual-balanced design, Modality-Occupancy imbalance, and context query count.

MOI	SOI	mAP	NDS
		70.7	72.6
✓		72.3	73.8
	✓	72.8	74.1
✓	✓	**73.2**	**74.5**

(a) Dual Balance

Components	mAP	NDS
Baseline-LC	70.7	72.6
+ Mo-Bal	71.7	73.3
+ GeoFuse	71.4	73.1
+ Mo-Bal (w/ GeoFuse)	**72.3**	**73.8**

(b) Modality Imbalance

# Query	mAP	NDS
16	72.3	73.6
32	72.5	74.0
64	**73.2**	**74.5**
128	70.2	72.5

(c) # Context Query

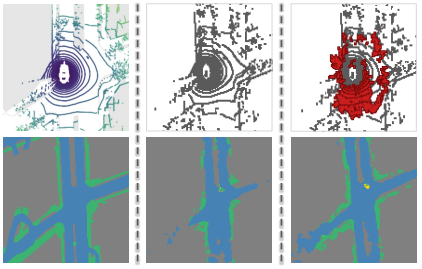
4.4 Ablation Studies

Dual Balance Ablation. Tab. 4(a) isolates the contributions of the MOI and SOI components. Starting from the 70.7 mAP baseline, mitigating MOI alone yields +1.6 mAP, while mitigating SOI alone yields a larger +2.1 mAP, indicating that semantic-occupancy imbalance is the more severe bottleneck in this

setting. Combining both achieves 73.2 mAP and 74.5 NDS—an additional +0.4 mAP beyond the better single component—confirming that the two imbalances interact and that joint mitigation unlocks complementary gains.

Impact of MOI Mitigation.

Tab. 4(b) breaks down the MOI component. Mo-Bal alone adds +1.0 mAP by enriching sparse regions with depth-guided pseudo-points, while GeoFuse alone adds +0.7 mAP by weighting modalities according to geometric reliability. Their combination yields +1.6 mAP and +1.2 NDS, exceeding the sum of individual gains and indicating a synergistic effect: balanced sampling provides GeoFuse with better-distributed features to weight. Figure 4 visualizes this effect—the pseudo-points generated by Mo-Bal (red) fill coverage gaps in the BEV map, producing more complete semantic predictions that in turn benefit Se-Bal's class-specific partitioning downstream.



(a) G.T. (b) *w/o* Ours (c) *w/* Ours

Fig. 4: Impact of Mo-Bal. Sampled image features (top) and the resulting BEV class predictions (bottom). Red points in (c) denote additional samples from Mo-Bal, producing more complete semantic maps.

Impact of SOI Mitigation. We analyze Se-Bal from three angles. First, Tab. 4(c) sweeps the number of context queries: 64 queries achieve the best balance at 73.2 mAP and 74.5 NDS, while 128 queries degrade to 70.2 mAP, suggesting that overly large query banks introduce redundant context that interferes with class-specific aggregation.

Table 5: Ablation of semantic-balanced enhancement.

Method	mAP	NDS
Baseline (Mo-Bal + GeoFuse)	72.3	73.8
+ Semantic-guided Context Attention	72.9	74.3
+ Semantic-guided Instance Attention	**73.2**	**74.5**

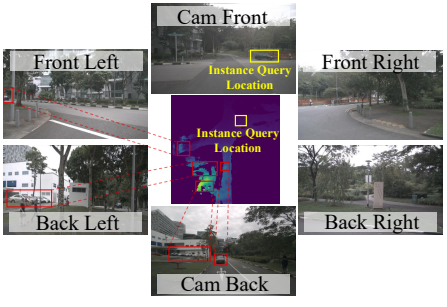
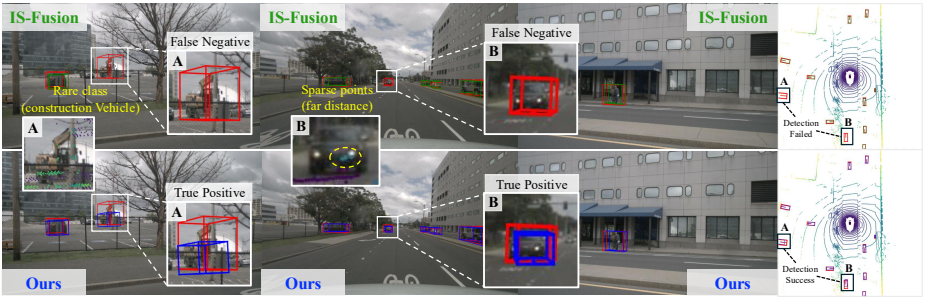


Fig. 5: Visualization of Semantic-guided attention map.

Second, Tab. 5 decomposes Se-Bal into its two stages: semantic-guided context attention contributes +0.6 mAP by routing context within semantic classes, and semantic-guided instance attention adds a further +0.3 mAP by refining instance-level features through same-class self-attention. Third, Fig. 5 visualizes the instance-query attention maps, where the yellow-box instance feature acts as the query and gathers information from BEV regions of the same semantic class that serve as keys and values, confirming that Se-Bal learns semantically coherent representations rather than attending indiscriminately to the entire scene.



(a) Camera Back Left (b) Camera Back (c) Camera Back Right (d) LiDAR Top

Fig. 6: Qualitative detection results. Red boxes denote ground-truth annotations, green boxes indicate baseline [37] predictions, and blue boxes indicate our predictions. The examples show that, even when LiDAR points are sparse or the object class is rare, our model mitigates false negatives by maximizing available information through comparisons with regions that share similar cues.

Qualitative Analysis Fig. 6 compares DB-Fusion with IS-Fusion [37] on challenging cases. In Fig. 6(a), DB-Fusion correctly detects a rare-class construction vehicle that IS-Fusion misses, illustrating how Se-Bal's class-specific context aggregation recovers weak foreground signals under SOI. In Fig. 6(b), DB-Fusion detects an object at long range despite severe LiDAR sparsity, demonstrating that Mo-Bal and GeoFuse maintain stable object evidence under MOI. Both cases are false negatives in the baseline, consistent with our thesis that the two imbalances are primary sources of missed detections.

5 Conclusion

We have presented DB-Fusion, a unified framework that jointly mitigates two structural bottlenecks of BEV-based LiDAR–camera fusion: Modality-Occupancy Imbalance (MOI) and Semantic-Occupancy Imbalance (SOI). Mo-Bal and GeoFuse close the cross-modal density gap and model geometry-dependent sensor reliability, while Se-Bal partitions the BEV map into semantic classes for class-consistent context aggregation and instance-scene interaction. We note that these modules are intentionally simple and handcrafted; our priority is to validate that the two imbalances are real bottlenecks, and stronger components would likely bring further gains. Experiments on nuScenes confirm this—mitigating both jointly yields 73.2 mAP on the validation set and 73.6 mAP on the test set with competitive NDS, with the largest gains emerging from their combination rather than either alone. Our analysis further reveals that the primary benefit lies in reducing false negatives, particularly in sparse point cloud conditions, while also promoting more reliable fusion that avoids over-reliance on a single modality, suggesting that balancing modality density and semantic context is a promising direction for improving overall detection quality in multimodal 3D detection.

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Supplementary Material

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A Overview

This supplementary material provides additional visualizations and analyses to complement our main paper.
It is organized as follows:

- Section B : Additional visualization of Mo-Bal
- Section C : Detailed corruption results for MOI analysis
- Section D : Visualization of GeoFuse's quality map
- Section E : Analysis of Se-Bal
- Section F : Additional qualitative detection results
- Section G : Limitations

B Additional visualization of Mo-Bal

To complement Sec. 3.2 and Fig. 4 in the main paper, we visualize in Fig. S1 how Modality-Balanced Feature Sampling (Mo-Bal) adaptively selects image features in BEV space. Fig. S1(a) shows gray points representing LiDAR-origin anchor samples that provide sparse but reliable 3D geometric evidence. Fig. S1(b) illustrates the sampling regions predicted by the depth-gating module, where red points indicate pseudo points generated by Mo-Bal that identify visible and geometrically plausible locations for additional feature extraction. Fig. S1(c) presents the merged feature set, combining both LiDAR-guided anchors and Mo-Bal pseudo points in the unified BEV representation.

Rather than naively lifting dense image features to all BEV cells, Mo-Bal employs the depth-gating mechanism to

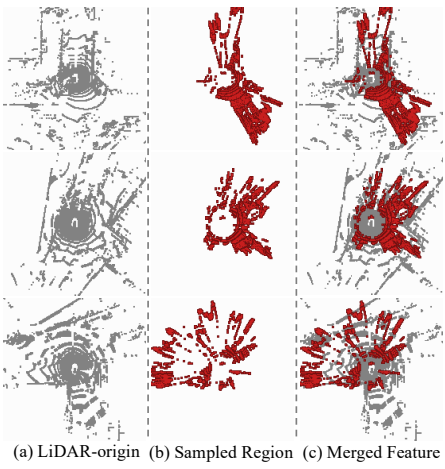

Fig. S1: Visualization of Modality-Balanced Feature Sampling. (a) LiDAR anchor points. (b) Mo-Bal pseudo points generated in depth-gated regions. (c) Combined BEV feature set of LiDAR anchors and pseudo points.

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restrict additional sampling to the regions highlighted in Fig. S1(b). As demonstrated in Fig. S1(b,c), the augmented samples are strategically concentrated around LiDAR-sparse yet informative areas, such as distant structures and partially occluded regions, while background cells without reliable depth evidence remain largely unsampled. This selective sampling strategy enables Mo-Bal to densify the image BEV representation in an adaptive, geometry-aware manner, effectively complementing LiDAR coverage without introducing noise from unreliable regions.

C Detailed corruption results for MOI analysis

To provide a more detailed analysis of Modality-Occupancy Imbalance (MOI) beyond the averaged results reported in the main paper, we provide in Table S1 the full per-severity results on nuScenes-C for Gaussian, Uniform, and Impulse corruptions. While the main paper reports the averages over five severity levels, the expanded results here make it possible to examine how each model behaves as the corruption strength increases.

Table S1: Full MOI analysis under camera-only corruptions across severity levels. Table 2 in the main paper reports results averaged over five severity levels, whereas here we provide the full per-severity results for each corruption type.

(a) Gaussian corruption

Method	Severity 1		Severity 2		Severity 3		Severity 4		Severity 5		Average	
	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS
IS-Fusion	70.16	72.44	68.40	71.43	65.69	69.90	62.84	68.40	61.32	67.66	65.68	69.99
MambaFusion	69.91	73.40	67.78	72.19	65.20	70.74	64.46	70.41	64.51	**70.46**	66.37	71.44
DB-Fusion (Ours)	**71.15**	**73.46**	**69.79**	**72.74**	**68.16**	**71.79**	**66.60**	**71.02**	**65.48**	70.45	**68.24**	**71.89**

(b) Uniform corruption

Method	Severity 1		Severity 2		Severity 3		Severity 4		Severity 5		Average	
	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS
IS-Fusion	71.53	73.18	70.61	72.70	68.96	71.77	66.65	70.46	63.69	68.85	68.29	71.39
MambaFusion	71.33	**74.23**	70.35	**73.67**	68.62	72.65	66.07	71.20	64.55	70.46	68.18	72.44
DB-Fusion (Ours)	**72.27**	74.05	**71.51**	73.62	**70.24**	**72.97**	**68.83**	**72.20**	**67.28**	**71.38**	**70.03**	**72.84**

(c) Impulse corruption

Method	Severity 1		Severity 2		Severity 3		Severity 4		Severity 5		Average	
	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS
IS-Fusion	69.82	72.27	67.68	71.04	66.02	70.09	63.23	68.59	61.61	67.75	65.67	69.95
MambaFusion	69.09	72.90	67.06	71.73	65.39	70.85	64.30	70.32	64.42	70.40	66.05	71.24
DB-Fusion (Ours)	**70.84**	**73.32**	**69.48**	**72.54**	**68.53**	**72.00**	**66.70**	**71.04**	**65.71**	**70.56**	**68.25**	**71.89**

As shown in Table S1, DB-Fusion consistently achieves the highest mAP across all severity levels for all three corruption types, while also maintaining

strong and competitive NDS throughout. The gains remain stable as the severity increases, indicating that DB-Fusion is less affected by camera corruption than the compared fusion baselines. These results further support the main-paper observation that alleviating MOI leads to improved fusion performance, especially when the visual input becomes degraded.

D Visualization of GeoFuse's quality map

To provide deeper insight into the GeoFuse module's behavior described in Sec. 3.3, we visualize in Fig. S2 the geometry-aware modality emphasis maps for a representative scene. As explained in the main paper, the module computes distance-based reliabilities $M_L(x, y)$ and $M_I(x, y)$ for LiDAR and image features through the distance-dependent weights w_{L} and w_{I}, along with an observation-quality term $M_{\mathrm{quality}}(x, y)$ derived from the angle-based quality modeling $q(i)$. For each BEV cell (x, y), we define the emphasis maps as

$$U_L(x, y) = M_{\mathrm{quality}}(x, y)\, M_L(x, y) \tag{S1}$$

$$U_I(x, y) = M_{\mathrm{quality}}(x, y)\, M_I(x, y) \tag{S2}$$

which represent the multiplicative gains applied to the LiDAR and image BEV features before fusion.

As illustrated in Fig. S2, the GeoFuse module does not follow a simplistic distance-only rule such as "near = LiDAR, far = image". Instead, the emphasis maps reveal a more nuanced pattern: LiDAR receives higher emphasis in regions where near-range geometry is reliable and point cloud returns are dense, while the image modality gains higher emphasis in distant or LiDAR-sparse areas where camera observations provide more informative cues.

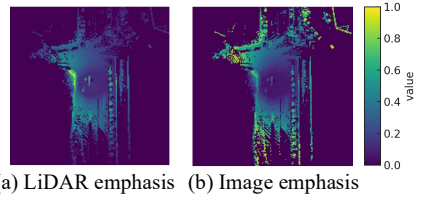

Fig. S2: Geometry-aware modality emphasis in BEV space. (a) LiDAR emphasis map U_L and (b) image emphasis map U_I, normalized to $[0, 1]$ and visualized as heatmaps. Brighter cells indicate higher modality emphasis.

Importantly, the angle-based quality term plays a crucial role in distinguishing reliable observations from unreliable ones based on elevation angles. Specifically, the quality modeling effectively suppresses emphasis on ground plane regions where steep elevation angles result in unreliable LiDAR measurements, while maintaining strong emphasis on object-occupied areas where points are observed near the horizontal plane. This visualization demonstrates that the proposed module exploits both distance and observation geometry to adaptively re-weight modalities in BEV space, enabling a more sophisticated fusion strategy that goes beyond simple range-based heuristics and effectively handles the complementary characteristics of each sensor.

E Analysis of Se-Bal

E.1 Se-Bal Segmentation Results

To complement Se-Bal module (Sec. 3.4) in the main paper, we visualize in Fig. S3 the BEV semantic segmentation maps that enable class-wise feature grouping in our Se-Bal module. We adopt a simple four-class discretization $S = \{\text{drivable}, \text{walkway}, \text{crosswalk}, \text{none}\}$ that explicitly models road-related regions where objects predominantly appear.

In this design, road markings such as lane dividers and stop lines are grouped into the drivable class rather than defining a separate lane category, as these elements share similar geometric and contextual characteristics with drivable surfaces. Although this segmentation head is not the primary focus of our framework and employs a relatively lightweight architecture, the predicted masks successfully separate the major semantic classes and provide reasonable class-wise partitions for the semantic-guided attention mechanism. These segmentation results demonstrate that even a simple class discretization is sufficient to enable effective semantic-based feature routing in the Se-Bal module.

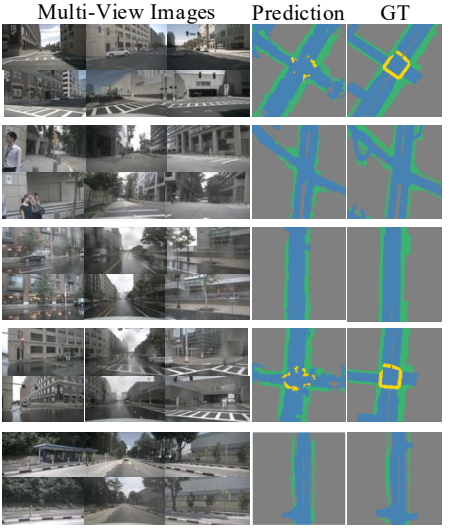

Fig. S3: BEV Segmentation Maps.

E.2 Visualization of Se-Bal attention map

To further illustrate the semantic-guided attention mechanism described in the main paper, we visualize in Fig. S4 additional examples of the semantic-guided instance-scene attention used in Se-Bal. For each highlighted instance, its feature serves as the key and value, while BEV features belonging to the same semantic class act as queries. The resulting attention weights are overlaid on the BEV grid and visualized from different surround-camera perspectives to demonstrate the spatial distribution of semantic-guided interactions.

As shown across multiple scenes, the attention responses are sharply concentrated within semantically consistent regions (e.g., drivable areas for vehicles, walkways for pedestrians), while unrelated background regions receive minimal activation. This spatial selectivity demonstrates that the semantic partitioning in Se-Bal effectively routes instance-scene interactions along class-specific regions and suppresses cross-class interference. By constraining attention to semantically relevant areas, Se-Bal enhances the discriminative power of BEV representations and enables more focused feature aggregation for each object category.

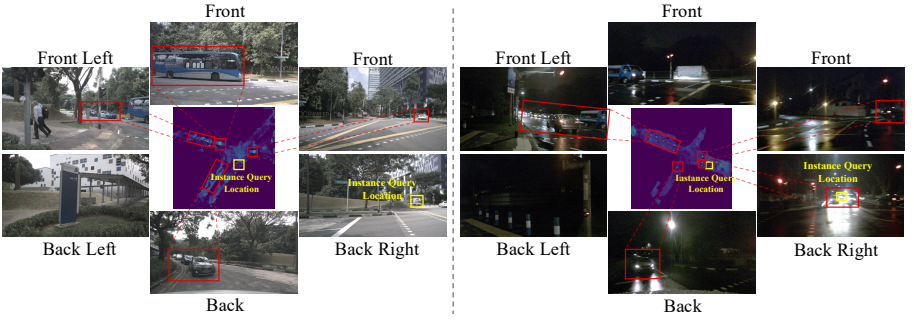

Fig. S4: Semantic-guided attention maps. For each highlighted instance (yellow box), we visualize the attention weights from BEV features of the same semantic class, overlaid on the BEV grid for different camera directions. Brighter regions indicate stronger semantic-guided instance–scene interaction around the query object.

F Additional qualitative detection results

To complement the quantitative detection metrics presented in the main paper, we provide in Fig. S5 qualitative examples that demonstrate our method's ability to reduce false negatives in challenging scenarios. Across diverse scenes, our detector consistently detects objects that are small, distant, or partially occluded, as well as targets located in regions with sparse LiDAR observations where conventional methods often fail. These visualization results highlight that our dual-balance fusion approach, which explicitly mitigates both Modality-Occupancy Imbalance and Semantic-Occupancy Imbalance, produces more complete 3D detection results. The improvements are particularly evident for distant vehicles and pedestrians in complex urban environments, where the complementary strengths of LiDAR and camera modalities are effectively leveraged through our geometry-aware and semantic-guided fusion strategies.

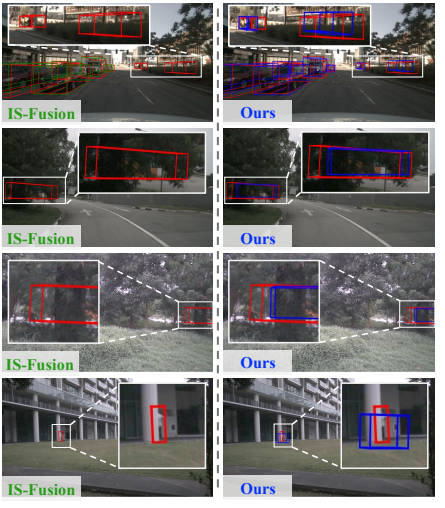

Fig. S5: Qualitative detection results and false negative reduction. We compare the predictions of IS-Fusion and our method on challenging scenes. Our detector recovers multiple objects, especially in distant or LiDAR-sparse regions, qualitatively demonstrating its ability to reduce false negatives.

G Limitations

While our proposed framework demonstrates strong performance in multi-modal 3D object detection, there are several aspects that could be further improved in future work.

Depth Estimation Module. Our current depth estimation module in Mo-Bal is relatively lightweight, relying on LiDAR points projected onto the image plane for supervision. As depth estimation is not the primary focus of our framework, there is room for improvement. A more sophisticated depth estimator, such as those trained on large-scale depth datasets or incorporating self-supervised learning techniques, would enable Mo-Bal to generate pseudo points that are more geometrically consistent with the underlying point clouds. With a stronger depth signal, the depth-gating parameters could be more strictly configured to further narrow the sampling region, allowing the module to focus on higher-quality image features while more effectively filtering out ambiguous or noisy depth predictions.

BEV Semantic Segmentation. The BEV semantic segmentation used in Se-Bal, while sufficient for our semantic-guided fusion, is not perfectly reliable in certain challenging scenarios. Complex intersections, heavily occluded regions, and visually ambiguous areas can lead to mis-segmentation of fine-grained structures and rare classes. Integrating more advanced segmentation architectures or incorporating temporal information from sequential frames could improve the robustness of semantic partitioning and further enhance the effectiveness of the Se-Bal module.

Despite these limitations, our experimental results demonstrate that even with these relatively simple implementations, the proposed dual-balance fusion framework achieves significant improvements over existing methods, suggesting substantial potential for further gains with more sophisticated depth estimation and segmentation components.